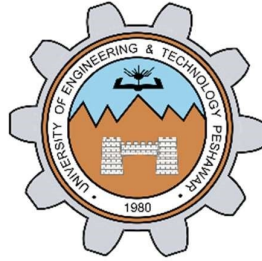


Designing and Implementing a Small Office Network Using Cisco Packet Tracer



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Computer communication and network

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“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Signature:

A handwritten signature in blue ink, appearing to read "Naeem Khan", with a long horizontal stroke extending to the right.

Submitted to:

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Mar 08, 2026

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1. Introduction:

Computer networks are an essential part of modern organizations, enabling efficient communication, resource sharing, and access to information. A well-designed Local Area Network (LAN) allows computers and networking devices within a limited geographical area, such as an office or campus, to communicate quickly and securely. An effective network infrastructure improves productivity, simplifies data sharing, and supports the future growth of an organization.

This lab focuses on designing and implementing a small office network for a startup company using Cisco Packet Tracer, a widely used network simulation tool. The proposed office consists of four departments: Management, Accounts, Sales, and IT Support, with a total of 11 computers connected through a central network infrastructure. The network is designed using a star topology, where all end devices are connected to a central switch, and a router acts as the default gateway for the network.

2. Problem Statement:

A startup company is establishing a new office consisting of four departments: Management (2 PCs), Accounts (3 PCs), Sales (4 PCs), and IT Support (2 PCs). The organization requires a reliable, cost-effective, and scalable Local Area Network (LAN) that enables seamless communication and resource sharing among all departments. The network must be designed within budget constraints while ensuring efficient connectivity, ease of management, and the ability to accommodate future expansion.

The primary challenge is to determine the most suitable network topology, select appropriate networking devices, choose the correct transmission media, and develop an effective IP addressing scheme. The network must then be implemented and configured in Cisco Packet Tracer, with successful connectivity verified through ping tests. Additionally, the proposed design should be evaluated in terms of performance, scalability, collision handling, and potential improvements, while comparing it with an alternative network design to justify the final solution.

3. Objectives:

The objectives of this project are:

- To analyze the networking requirements of a small office environment and identify an appropriate LAN solution.
- To design an efficient and cost-effective LAN topology for connecting the Management, Accounts, Sales, and IT Support departments.

- To select suitable networking devices, transmission media, and an IP addressing scheme based on the given requirements.
- To implement the proposed network in Cisco Packet Tracer by configuring the router, switch, and end devices.
- To verify end-to-end network connectivity using ping tests and ensure successful communication between all devices.
- To evaluate the network design in terms of performance, collision handling, scalability, and future expansion.
- To compare the proposed network with an alternative design and justify the selected solution based on cost, performance, scalability, and security.
- To develop practical skills in network design, configuration, testing, and troubleshooting using Cisco Packet Tracer.

4. Network Requirements:

The proposed network is designed to provide reliable and efficient communication for a startup company consisting of four departments: Management, Accounts, Sales, and IT Support. The network must support 11 computers, enable seamless resource sharing, and ensure stable connectivity while maintaining a cost-effective and scalable infrastructure. A star topology is selected to simplify network management, improve reliability, and facilitate future expansion.

The network utilizes a Cisco 2911 router as the default gateway and a Cisco Catalyst 2960 switch to connect all end devices. Static IPv4 addressing is implemented using the 192.168.10.0/24 network, and Copper Straight-Through Ethernet cables are used to establish connections between the router, switch, and computers. This design provides a simple, reliable, and scalable solution that meets the organization's current requirements while supporting future growth.

Table 4.1: Network Requirements Summary

Requirement	Specification
Network Type	Local Area Network (LAN)
Departments	Management, Accounts, Sales, IT Support
Total PCs	11

Requirement	Specification
Topology	Star Topology
Router	Cisco 2911
Switch	Cisco Catalyst 2960
IP Addressing	Static IPv4
Network Address	192.168.10.0/24
Transmission Medium	Copper Straight-Through Cable

5. Network Design:

5. Network Design

The network is designed using a Star Topology, where all end devices are connected to a central switch. This topology is chosen because it is simple to implement, easy to manage, scalable, and provides better fault isolation. If one device fails, the rest of the network continues to operate without interruption.

A Cisco 2911 router is used as the default gateway for the network. The router connects to the Cisco 2960 switch, which operates at Layer 2 and connects all departments' computers. All devices are configured in the same subnet (192.168.10.0/24) to allow communication across the entire network.

This design satisfies the requirement of reliable communication, resource sharing, and future expansion with minimal changes.

5.1 Department-wise Distribution of PCs

Department	Number of PCs
Management	2
Accounts	3
Sales	4
IT Support	2
Total	11

Design Summary

- Star topology with a central switch.
- Cisco 2911 router acts as the default gateway.
- Cisco 2960 switch connects all end devices.
- All devices use the 192.168.10.0/24 network.
- The design is scalable, reliable, and easy to manage.

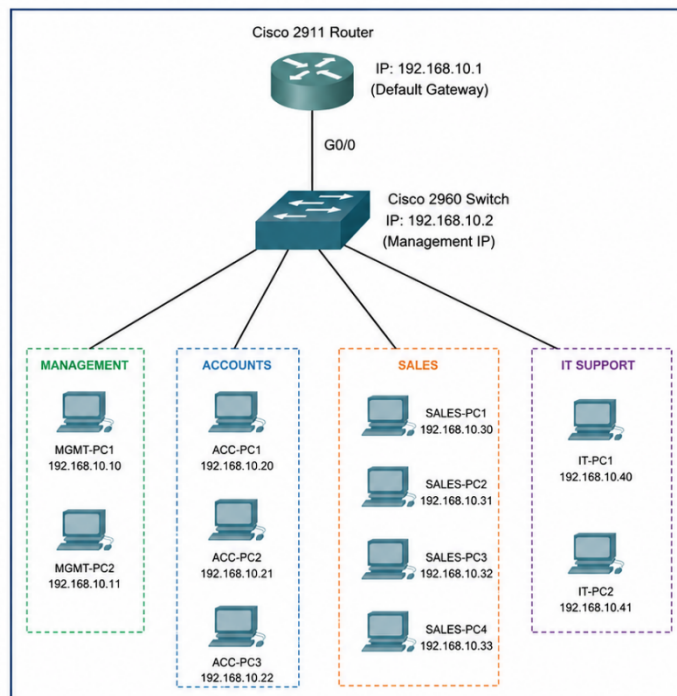


Figure 5.1: Network Topology Diagram

6. Device Selection:

The selection of networking devices plays a vital role in ensuring reliable communication, efficient performance, and future scalability. The devices used in this network were selected based on the requirements of the proposed office environment, considering factors such as functionality, cost-effectiveness, ease of configuration, and expansion capability.

Table 6.1: Selected Network Devices

Device	Model	Quantity	Purpose
Router	Cisco 2911	1	Acts as the default gateway and enables communication between the local network and external networks.
Switch	Cisco Catalyst 2960	1	Connects all end devices within the LAN and forwards data efficiently between them.
Personal Computer (PC)	Generic PC	11	Represents end-user devices used by employees in different departments.
Ethernet Cable	Copper Straight-Through	12	Establishes physical connections between the router, switch, and computers.

Device Justification:

Cisco 2911 Router:

The Cisco 2911 router was selected to function as the default gateway for the network. It provides reliable routing capabilities, supports IPv4 addressing, and offers sufficient performance for a small office environment while allowing future network expansion.

Cisco Catalyst 2960 Switch:

The Cisco 2960 switch was chosen as the central networking device because it provides high-speed Layer 2 switching, supports multiple Fast Ethernet ports, and offers reliable communication among all connected devices. Its available ports also accommodate future additions to the network.

Personal Computers:

A total of eleven personal computers were deployed to represent the workstations of employees in the Management, Accounts, Sales, and IT Support departments. Each

computer was assigned a unique static IP address to ensure reliable communication within the LAN.

CopperStraight-Through Ethernet Cables:

Copper Straight-Through cables were used to connect the router to the switch and the switch to each PC. This cable type is the standard choice for connecting different types of networking devices and ensures stable data transmission.

7. Cable Selection:

The selection of an appropriate transmission medium is essential for ensuring reliable and efficient communication within a Local Area Network (LAN). Since the proposed network consists of a router, a switch, and multiple end-user computers, Copper Straight-Through Ethernet cables were selected for all physical connections. This cable type is the standard choice for connecting different types of networking devices, such as a router to a switch and a switch to a computer.

Copper Straight-Through cables provide stable data transmission, are cost-effective, and are widely used in Ethernet-based LANs. Their compatibility with Fast Ethernet interfaces makes them suitable for small office environments, ensuring reliable connectivity while simplifying installation and maintenance.

Table 7.1: Cable Selection

Connection	Cable Type	Reason for Selection
Router → Switch	Copper Straight-Through	Used to connect different networking devices and provide gateway connectivity.
Switch → Management PCs	Copper Straight-Through	Provides reliable Ethernet connectivity between the switch and end devices.
Switch → Accounts PCs	Copper Straight-Through	Ensures stable communication within the LAN.
Switch → Sales PCs	Copper Straight-Through	Supports efficient data transmission between end devices and the switch.
Switch → IT Support PCs	Copper Straight-Through	Enables reliable and high-speed LAN communication.

Justification:

Copper Straight-Through Ethernet cables were selected because they are the recommended medium for connecting unlike devices in Ethernet networks. They offer dependable performance, easy deployment, and low implementation cost, making them an appropriate choice for the proposed small office network. Their use ensures stable communication between all network devices while supporting the project's requirements for reliability, simplicity, and scalability.

8. IP Addressing Scheme:

An appropriate IP addressing scheme is essential for ensuring efficient communication and effective network management. In the proposed network, a private IPv4 Class C address range (192.168.10.0/24) is used to assign unique IP addresses to all network devices. This addressing scheme provides sufficient host addresses for the current office environment while allowing future expansion without requiring major network modifications.

Static IP addressing is implemented for all devices because the network consists of a limited number of computers and does not require dynamic address allocation. Each device is assigned a unique IP address within the same subnet to facilitate seamless communication. The Cisco 2911 router is configured as the default gateway (192.168.10.1), while the Cisco Catalyst 2960 switch is assigned a management IP address (192.168.10.2) to enable administrative access. All end devices use the subnet mask 255.255.255.0, and 8.8.8.8 is configured as the DNS server.

Table 8.1: IP Addressing Scheme

Device	IP Address	Subnet Mask	Default Gateway
Router (R1)	192.168.10.1	255.255.255.0	—
Switch (SW1)	192.168.10.2	255.255.255.0	192.168.10.1
MGMT-PC1	192.168.10.10	255.255.255.0	192.168.10.1
MGMT-PC2	192.168.10.11	255.255.255.0	192.168.10.1
ACC-PC1	192.168.10.20	255.255.255.0	192.168.10.1
ACC-PC2	192.168.10.21	255.255.255.0	192.168.10.1
ACC-PC3	192.168.10.22	255.255.255.0	192.168.10.1
SALES-PC1	192.168.10.30	255.255.255.0	192.168.10.1

Device	IP Address	Subnet Mask	Default Gateway
SALES-PC2	192.168.10.31	255.255.255.0	192.168.10.1
SALES-PC3	192.168.10.32	255.255.255.0	192.168.10.1
SALES-PC4	192.168.10.33	255.255.255.0	192.168.10.1
IT-PC1	192.168.10.40	255.255.255.0	192.168.10.1
IT-PC2	192.168.10.41	255.255.255.0	192.168.10.1

IP Addressing Summary

Parameter	Value
Network Address	192.168.10.0/24
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	8.8.8.8
Addressing Method Static IPv4	

This IP addressing scheme ensures that every network device has a unique address within the same subnet, enabling reliable communication while maintaining a simple and scalable network architecture suitable for the proposed small office environment.

9. Network Implementation:

The proposed Local Area Network (LAN) was implemented in Cisco Packet Tracer by following a structured approach that included network topology creation, device interconnection, IP address configuration, and connectivity verification. A Cisco 2911 router and a Cisco Catalyst 2960 switch were deployed as the core networking devices, while 11 personal computers were connected to the switch using Copper Straight-Through Ethernet cables.

Initially, the network topology was designed using a star topology, where all end devices were connected to the central switch. The router was connected to the switch through the GigabitEthernet0/0 interface and configured with the IP address 192.168.10.1/24, serving

as the default gateway for the network. The switch was assigned the management IP address 192.168.10.2/24, enabling basic network administration.

Each workstation was configured with a unique static IPv4 address within the 192.168.10.0/24 network. The subnet mask 255.255.255.0 and the default gateway 192.168.10.1 were configured on every computer to ensure communication within the local network. After completing the device configuration, connectivity was verified using the ping utility. Successful responses between the router and all end devices confirmed that the network was functioning correctly with 0% packet loss, indicating proper configuration and reliable communication.

Implementation Steps

1. Designed the network topology in Cisco Packet Tracer using a star topology.
2. Added one Cisco 2911 router, one Cisco Catalyst 2960 switch, and eleven PCs.
3. Connected the router to the switch and all PCs to the switch using Copper Straight-Through Ethernet cables.
4. Configured the router interface with the IP address 192.168.10.1/24.
5. Assigned the management IP address 192.168.10.2/24 to the switch.
6. Configured static IPv4 addresses on all end-user computers.
7. Verified end-to-end connectivity using the ping command.
8. Saved the configurations of the router and switch to ensure persistence after reboot.

Figures:

Figure 9.1: Final Network Topology:

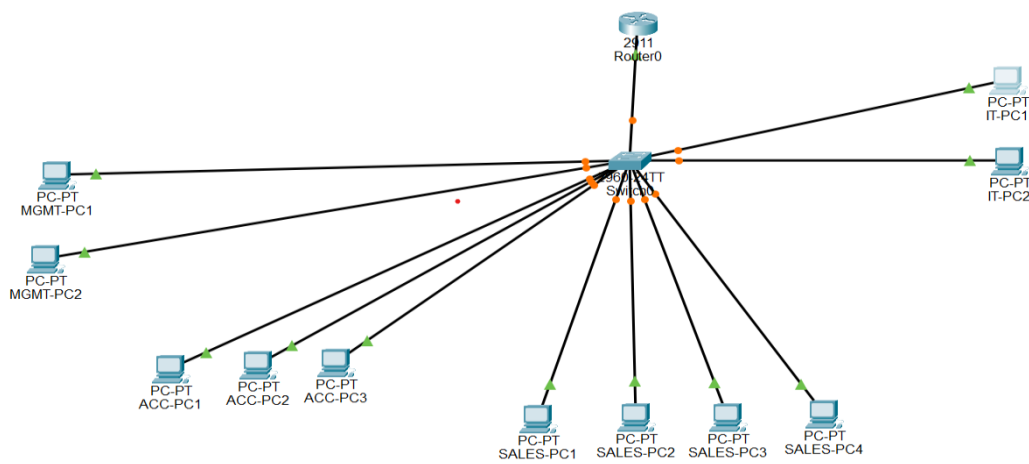


Figure 9.2: Router Configuration:

```
--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#no ip domain-lookup
R1(config)#interface gigabitEthernet0/0
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down

R1(config-if)#exit
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#
```

Figure 9.3: Switch Configuration:

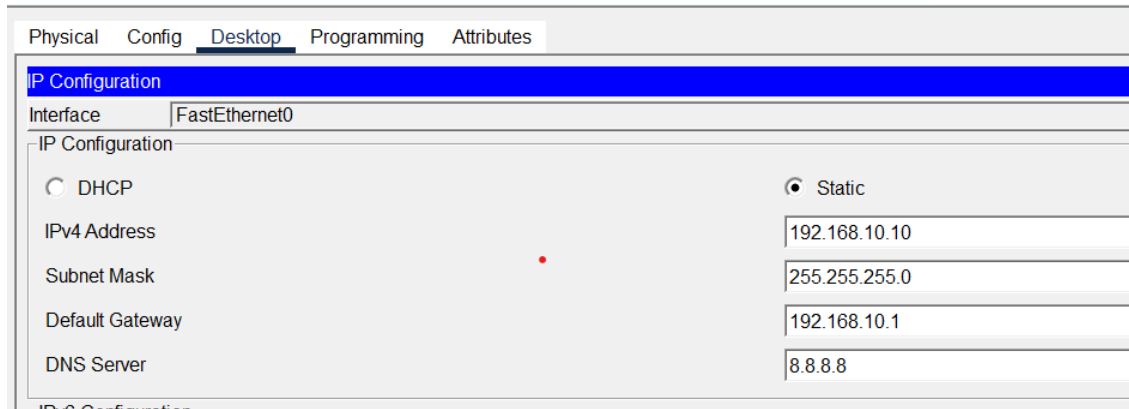
```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW1
SW1(config)#interface vlan 1
SW1(config-if)#ip address 192.168.10.2 255.255.255.0
SW1(config-if)#no shutdown

SW1(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

SW1(config-if)#exit
SW1(config)#ip default-gateway 192.168.10.1
SW1(config)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
SW1#
```

Figure 9.4: PCs Configuration:



And we repeat this process for all the PCs.

Figure 9.5: Successful Ping Test Showing 0% Packet Loss

```
C:\>ping 192.168.10.20

Pinging 192.168.10.20 with 32 bytes of data:

Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

10. Performance Analysis:

The implemented network was evaluated based on its connectivity, reliability, scalability, and overall performance. The use of a star topology, together with a Cisco 2911 router and a Cisco Catalyst 2960 switch, provides a stable and efficient Local Area Network (LAN) suitable for the operational requirements of the startup company. Connectivity tests performed using the ping utility confirmed successful communication between all devices, with 0% packet loss, indicating that the network was configured correctly and is capable of supporting reliable data transmission.

10.1. Network Performance:

The network demonstrated efficient communication among all connected devices. Since all computers are connected through a dedicated switch, data frames are forwarded only to the intended destination, reducing unnecessary network traffic and improving overall performance. The successful ping results verified that all hosts could communicate with the router and with one another without noticeable delay.

10.2. Collision Analysis:

The Cisco Catalyst 2960 switch significantly reduces network collisions by creating a separate collision domain for each connected device. Unlike a hub, which broadcasts data to all connected devices, the switch forwards data only to the destination port based on its MAC address table. This improves bandwidth utilization, minimizes collisions, and increases communication efficiency. Furthermore, the use of full-duplex communication eliminates collisions during simultaneous data transmission and reception.

10.3. Scalability:

The proposed network is designed with future expansion in mind. The Cisco Catalyst 2960 switch provides additional unused ports that can accommodate more computers or networking devices without requiring major modifications to the existing infrastructure. The selected IP addressing scheme (192.168.10.0/24) also provides sufficient address space to support additional hosts if the organization grows.

10.4. Future Improvements:

Although the current network satisfies the requirements of a small office, several enhancements could improve its performance and security in the future:

- Implement VLANs to separate departmental traffic and improve network security.
- Deploy a DHCP server to automate IP address assignment and simplify network administration.
- Install a firewall to protect the network from unauthorized access and external threats.
- Add a wireless access point to support wireless devices such as laptops and smartphones.
- Introduce a file server and network printer to enable centralized resource sharing and improve operational efficiency.

11. Alternative Design:

Although the proposed network satisfies the current requirements of the startup company, an alternative design can be considered to improve scalability, performance, and network security. The alternative design involves dividing the network into multiple Virtual Local Area Networks (VLANs), where each department (Management, Accounts, Sales, and IT Support) operates within its own logical network. Inter-VLAN communication can be achieved through the Cisco 2911 router using router-on-a-stick configuration.

Compared with the implemented single-LAN design, the VLAN-based approach reduces unnecessary broadcast traffic, enhances security by isolating departmental communications, and simplifies network management. Additionally, a second switch can be deployed to support future expansion, ensuring that the network can accommodate additional employees and devices without affecting performance.

However, the proposed alternative requires more advanced configuration, including VLAN creation, trunk port configuration, and inter-VLAN routing. It also increases implementation cost due to additional networking equipment and administrative complexity. Considering the limited size and budget of the startup company, the implemented single-switch, single-subnet design remains the most practical and economical solution while still providing reliable communication and adequate scalability.

Table 12.1: Comparison of Network Designs

Feature	Implemented Design	Alternative Design
Network Topology	Star Topology	Star Topology with VLANs
Number of Switches	1	2
Network Segmentation	Single LAN	Multiple VLANs
Cost	Low	Moderate
Configuration Complexity	Low	High
Scalability	Good	Excellent
Security	Basic	Enhanced through VLAN isolation
Broadcast Traffic	Higher	Reduced
Suitability	Small Office	Medium to Large Organizations

Justification:

Considering the organization's size, budget, and operational requirements, the implemented network design is the most appropriate solution. It provides reliable connectivity, simple administration, and sufficient scalability for future growth while maintaining low implementation and maintenance costs. Although the VLAN-based design offers improved security and performance, its additional complexity and cost are not essential for the current needs of the startup company.

12. Reflection:

This project provided valuable practical experience in designing, implementing, and testing a Local Area Network (LAN) using Cisco Packet Tracer. It strengthened my understanding of fundamental networking concepts, including network topology, IP addressing, router and switch configuration, and connectivity verification. Although these concepts were introduced in lectures, implementing them in a simulated environment significantly improved my confidence in applying theoretical knowledge to a practical scenario.

During the implementation process, I encountered several challenges, including selecting the appropriate network devices, configuring router and switch interfaces, assigning correct IP addresses, and ensuring successful communication between all devices. By carefully analyzing the network configuration, verifying interface settings, and performing connectivity tests using the ping command, I was able to identify and resolve these issues effectively. This experience enhanced my troubleshooting and problem-solving skills.

The project also emphasized the importance of proper network planning before implementation. Designing an organized IP addressing scheme, selecting suitable networking devices, and choosing an appropriate topology contributed significantly to the successful operation of the network. Furthermore, evaluating the network's scalability and considering future improvements, such as VLAN implementation and automated IP address allocation through DHCP, provided a broader understanding of how professional networks are designed and managed.

Overall, this lab enhanced my practical knowledge of computer networking and demonstrated the relationship between theoretical concepts and real-world implementation. The experience has improved my technical skills in network configuration, testing, and documentation, while also reinforcing the importance of systematic planning, careful configuration, and effective troubleshooting in building reliable and scalable computer networks.

13. Conclusion:

This project successfully achieved the objective of designing and implementing a reliable Local Area Network (LAN) for a startup company using Cisco Packet Tracer. A star topology was selected to provide a simple, efficient, and scalable network architecture. The network was implemented using a Cisco 2911 router, a Cisco Catalyst 2960 switch, and 11 end-user computers, all configured with a structured static IPv4 addressing scheme.

The router, switch, and end devices were successfully configured, and network connectivity was verified through ping tests, which confirmed successful communication among all devices with 0% packet loss. These results demonstrate that the proposed network meets the communication requirements of the organization while providing reliable performance and efficient data transmission.

The project also highlighted the importance of proper network planning, appropriate device selection, and systematic IP addressing in developing a functional and manageable network infrastructure. Furthermore, the performance analysis and alternative design evaluation demonstrated that the implemented network offers a practical balance between cost, simplicity, and scalability, while identifying future enhancements such as VLAN implementation, DHCP services, and improved network security.

Overall, this project provided valuable practical experience in network design, configuration, testing, and troubleshooting. It successfully bridged theoretical networking concepts with real-world implementation and demonstrated the ability to design a cost-effective, reliable, and scalable network suitable for a small office environment.